

Intermediate Report: Vision-Based Shooting Target Classification System

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1 Introduction

This work will develop a computer vision system for automatically scoring shooting impacts on ISSF paper targets. The project belongs to the field of Electrical and Computer Engineering, in the areas of image processing and machine learning.

The motivation behind this project comes from the need for a low-cost automatic scoring system in the shooting sport. Current systems that deliver accurate and automatic scores typically rely on special sensors and become expensive.

The main goal is to design a reliable, low-cost, real-time scoring system that can detect, classify, and score bullet impacts based on images captured by a webcam. The resulting system will be installed in a shooting club and will improve the user experience during training and competitions.

2 Background

2.1 Problem Statement

This project aims to develop a system capable of automatically scoring shooting targets for two ISSF disciplines: the *10 m air rifle* and the *10 m air pistol* events. The corresponding paper targets are shown in Figure 1. Each target consists of ten concentric scoring rings numbered from 1 to 10. Scoring is based on the distance between the bullet impact center and the geometric center of the target (the “10.9 ring”), following the official ISSF regulations [1].



(a) Air rifle target



(b) Air pistol target

Figure 1: ISSF 10 m targets used in this study.

According to the ISSF rulebook [1], the decimal scoring value S for a given shot is computed from the radial distance d (in mm) between the bullet's center and the target center using a linear scoring model:

$$S = 10.9 - k \cdot d, \quad (1)$$

where k is a constant that depends on the target type and determines the rate of score decrease per millimeter of deviation. For smaller targets such as those used in the air rifle discipline, the value of k is larger, reflecting a higher scoring sensitivity to positional errors.

Scoring targets for the 10 m air rifle category (Figure 1a) will therefore be more challenging than for the 10 m air pistol category (Figure 1b), due to the reduced target dimensions and the smaller ring spacing. Accurate image-based measurement is required to achieve sub-millimeter precision.

The camera placement in the envisioned system is illustrated in Figure 2. The camera will be positioned laterally, viewing the target at an oblique angle relative to the shooting line. As a result, the acquired images will have perspective distortion that must be compensated through image rectification.

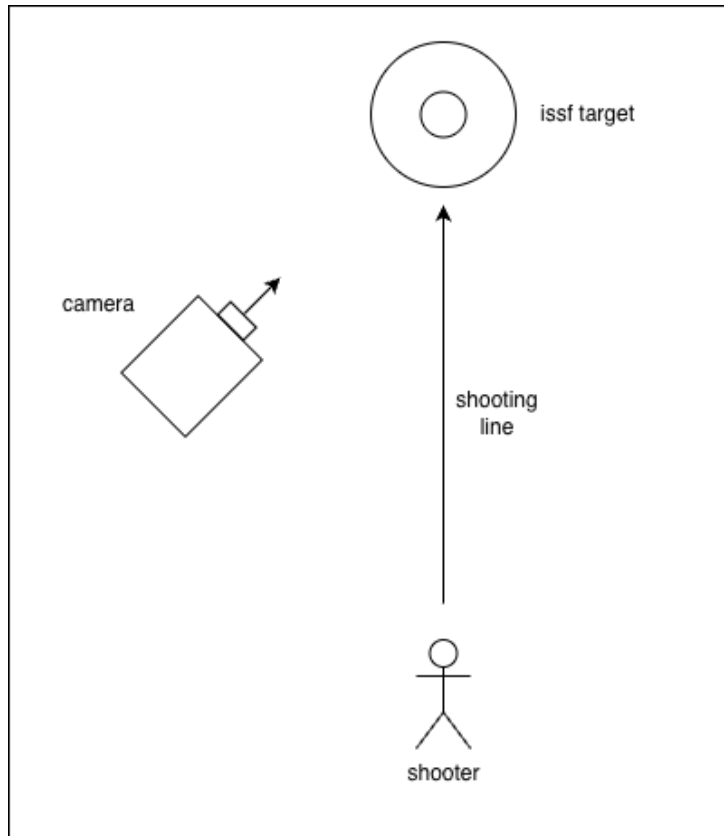


Figure 2: The camera is placed on the side of the shooting line, looking diagonally at the target.

During operation, the system will encounter targets containing both isolated and overlapping bullet impacts. As shown in Figure 3, overlapping bullets produce merged, non-circular contours, which complicates segmentation and localization. A robust detection and scoring algorithm must therefore be developed to accurately identify bullet centers and compute the corresponding scores under these challenging conditions.

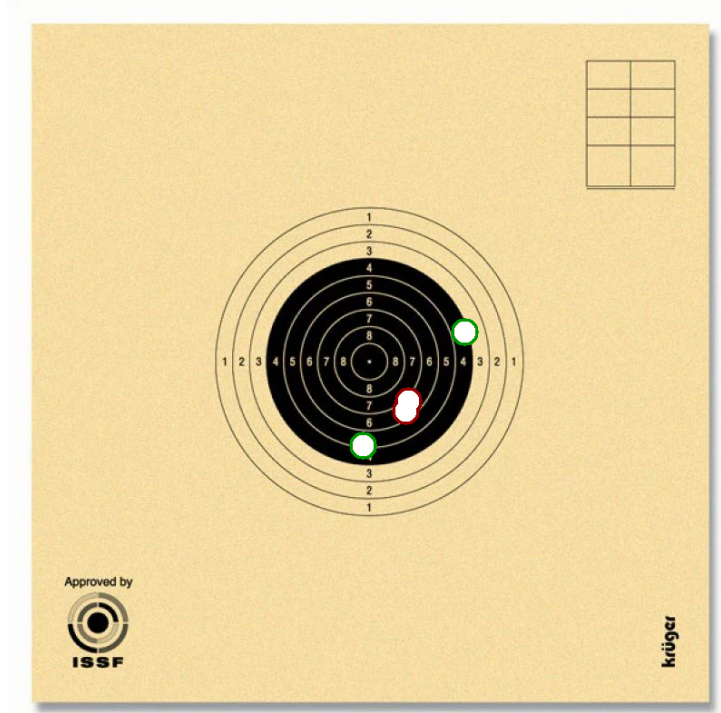


Figure 3: target with bullets.

2.2 Classical Image Processing Approaches

Early development of automatic scoring systems used classical image processing techniques. A typical pipeline used starts with image acquisition, where the perspective and camera distortion are removed. Then a pre-processing stage removes noise from the image, using some type of blur (low-pass filter) on the image adequate for the type of noise. After that, a local or global threshold is used to distinguish the ROI (region of interest) from the background. The ROI in this case are the bullet holes and the target center. The contours of the binary mask that constitutes the ROI are extracted. Finally, a circle regression is computed for each detected contour, finding the bullet hole centers and radii.

When multiple bullet holes overlap, this method may fail to separate them. To address this problem, algorithms like the *watershed transform* were developed. Liu *et al.* [2] proposed an improvement of the watershed algorithm that successfully separated overlapping bullet holes in controlled conditions.

Despite these advances, classical image processing remains highly sensitive to lighting variations, target texture, and background contrast. Each new shooting setup typically requires manual adjustment of thresholds and filtering parameters. Consequently, these methods often lack generalization and robustness when applied to different cameras, lighting environments, or target types.

2.3 Machine Learning and Deep Learning Approaches

Because of the drawbacks of classical approaches, computer vision research has shifted to data-driven learning using convolutional neural networks (CNNs) (as in figure 4). A CNN processes an image as a two-dimensional matrix of pixel intensities. Through successive convolution layers, it extracts increasingly abstract features—edges, textures, and finally object-level patterns. The deeper layers feed into a fully connected neural network (the “head”), which performs the final classification or detection task.

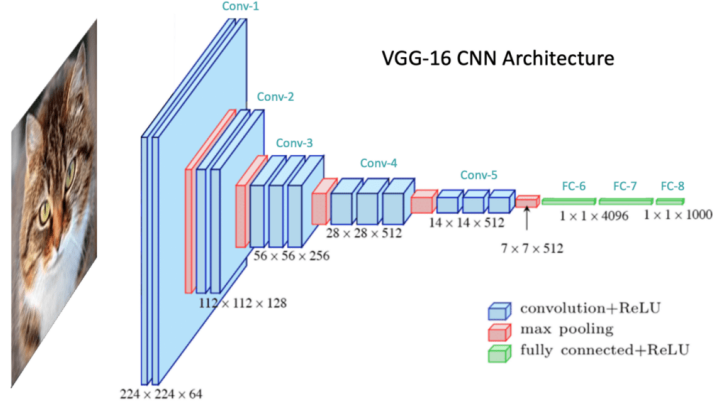


Figure 4: Conceptual representation of a convolutional neural network (CNN). Early layers extract simple visual features, while deeper layers represent higher-level patterns used for object detection or classification.

Rich datasets were used to create powerful pretrained models that are capable of extracting "good features" (e.g. edges and connected components) from a broader spectrum of images. These models are more robust to lighting conditions and represent a more general computer vision solution.

Deep learning-based methods such as Faster R-CNN, RetinaNet, and YOLO (You Only Look Once) architectures have demonstrated excellent performance in general object detection tasks. These models combine feature extraction, region proposal, and classification in an end-to-end pipeline that can generalize across diverse visual conditions. Their strength lies in their ability to automatically learn relevant spatial features, reducing dependence on manually tuned thresholds or lighting adjustments.

In the context of target scoring, Butt *et al.* [3] evaluated several state-of-the-art object detection networks — including RetinaNet, Faster R-CNN, and the anchor-free YOLOv8 family — on a dataset of paper targets. Their experiments showed that YOLOv8 achieved the highest detection accuracy (mAP50 of 96.7%) with real-time inference speed.

The downside to machine learning approaches is the amount of labeled training data and computational power necessary to train the model. The accuracy of the detection depends heavily on the quality and completeness of the training dataset. Thanks to the pretrained models that used millions of images to train on, the size of the dataset necessary for training the high level part of the neural network is much smaller, but still should be a few thousand images large. Techniques like data augmentation will come in handy to make the dataset more complete.

Despite the long time it takes to train a neural network, once the neural network is trained, the inference time is usually very fast, usually from 1ms to 10ms.

3 Conclusions

The machine learning approach looks very promising for real-time classification of bullet holes in a target. The precision of the position of the target's center and of the bullet hole's centers will have to be high. Eventually, a hybrid approach starting with a neural network inference to find bounding boxes and then a classical approach algorithm to accurately determine the relevant coordinates of bullets and bull's eye will be implemented.

After having an accurate image acquisition and scoring algorithm, we will create a database where athletes and their results are saved. Also, a virtual target will be displayed with the real-time information of the position of the bullets.

References

- [1] International Shooting Sport Federation, *ISSF Technical Rules: Rifle, Pistol and Shotgun*, 2023rd ed., Munich, Germany, 2023, sections 6.3.3–6.3.4, Annex: Rules for Paper Target Scoring. [Online]. Available: <https://www.issf-sports.org/theissf/rules.ashx>
- [2] P. Liu, Y. Chen, S. Kong, H. Wang, S. Fu, and J. Wang, “Overlapping bullet hole detection based on improved watershed algorithm,” in *Proceedings of the 2023 8th International Conference on Signal and Image Processing (ICSIP)*. Wuxi, China: IEEE, July 2023, pp. 136–140.
- [3] M. Butt, N. Glas, J. Monsuur, R. Stoop, and A. de Keijzer, “Application of yolov8 and detectron2 for bullet hole detection and score calculation from shooting cards,” *Ai*, vol. 5, no. 1, pp. 72–90, 2023.